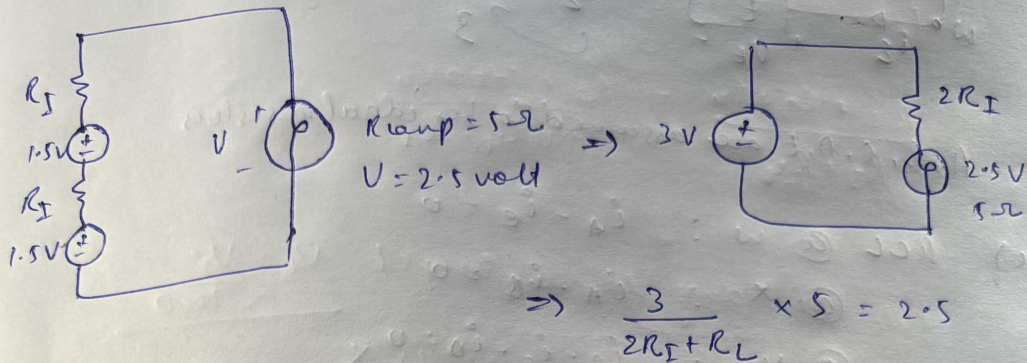


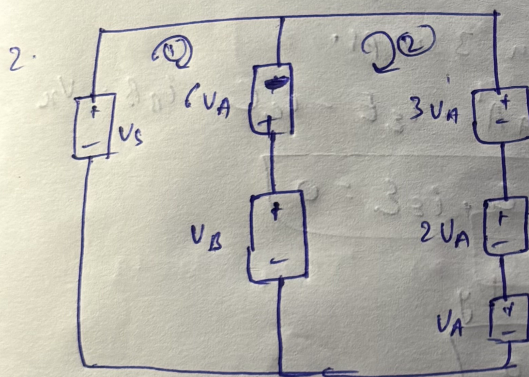
1. Abstracted equivalent circuit will be as follows



$$2R_I + 5 = 6$$

$$R_I = 0.5 \Omega$$

2. Internal Resistance of each battery = 0.5Ω



KVL in loop 1

$$V_S + 6V_A - V_B = 0$$

$$V_B = V_S + 6V_A \quad \text{--- (I)}$$

KVL in loop 2

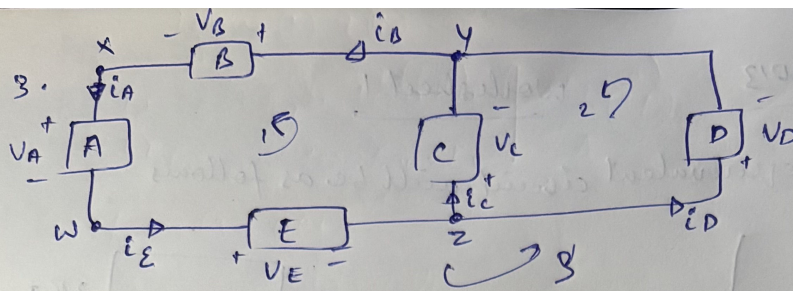
$$V_B - 6V_A - 3V_A - 2V_A - V_A = 0$$

$$V_B = 12V_A \quad \text{--- (II)}$$

from (I) and (II)

$$V_S = 6V_A$$

$$\Rightarrow \boxed{V_A = \frac{V_S}{6}} \quad \& \quad \boxed{V_B = 2V_S}$$



A, B, C, D, E are assumed to be ideal resistors

c) KCL @ w : $i_A - i_E = 0$

@ x : $i_A - i_B = 0$

@ y : $i_B - i_C - i_D = 0$

@ z : $i_E - i_C - i_D = 0$

Constructing in matrix we get rank = 3

$\therefore$ 3 linearly independent KCL eqn in this network

b) Similarly writing KVL in 3 loops.

KVL @ loop 1 : $V_x - i_A A - i_E E - i_C C - i_B B = V_x$

$$\Rightarrow i_A A + i_B B + i_C C + i_E E = 0$$

@ loop 2 : $V_y + i_C C - i_D D = V_y$

$$\Rightarrow i_C C - i_D D = 0$$

@ loop 3 : $V_x - i_A A - i_E E - i_D D - i_B B = V_x$

$$\Rightarrow i_A A + i_B B + i_D D + i_E E = 0$$

Rank of Matrix = 2 for these eqn

$\therefore$ 2 linearly independent KVL eqn for this network

d) for this network:

KCL eqⁿ are :

$$\begin{aligned} i_A &= i_E \\ i_A &= i_B \\ i_B &= i_C + i_D \end{aligned}$$

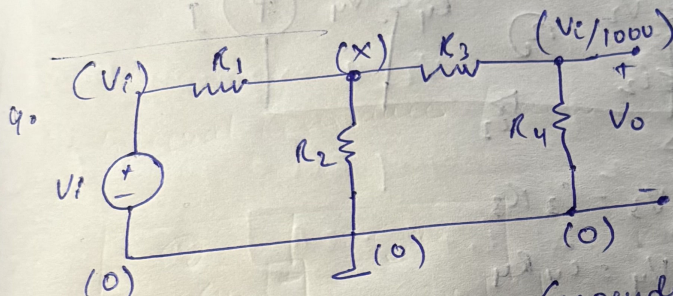
KVL eqⁿ are :

$$\begin{aligned} i_C C &= i_D D \\ i_A A + i_B B + i_C C + i_E E &= 0 \end{aligned}$$

e) $i_A = i_E = i_B = 2A$; $i_C = 1A$; $i_D = 1A$

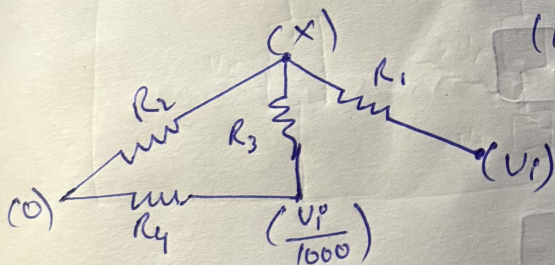
f) $V_C = 2 \text{ volt} = V_D$; $V_B = 1V$; $V_E = 1V$; $V_A = -4V$

$$\begin{cases} i_x X = V_x \\ X = A, B, C, D, E \end{cases}$$



$$V_o = \frac{V_i}{1000}$$

Grounding the node below R_2 and rearranging the circuit. (In bracket, potential is mentioned w.r.t. ground)



$$V_i - i_1 R_1 - i_2 R_2 = 0 \Rightarrow V_i = i_1 R_1 + i_2 R_2 \quad (I)$$

$$0 + R_4 (i_1 - i_2) = \frac{V_i}{1000} \quad (II)$$

from (I) & (II)

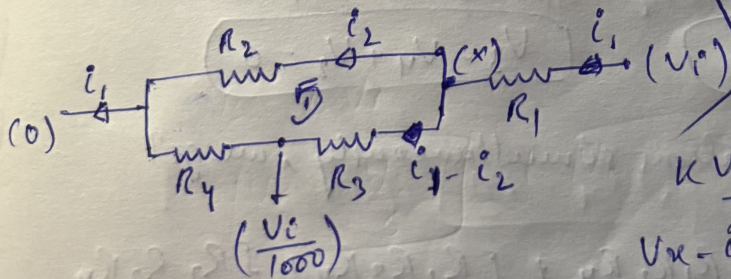
$$i_1 (1000 R_4 - R_1) = i_2 (R_2 + 1000 R_4)$$

KVL in loop 1

$$V_x - i_2 R_2 + R_4 (i_1 - i_2) + R_3 (i_3 - i_2) = V_o$$

$$i_1 R_4 + i_1 R_3 = i_2 R_2 + i_2 R_4 + i_2 R_3$$

$$\frac{i_1}{i_2} = \frac{R_2 + R_3 + R_4}{R_3 + R_4} \quad (3)$$



from (1) & (1)

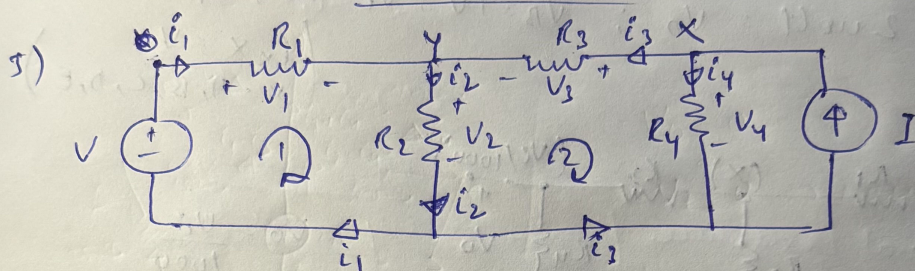
$$\frac{i_1}{i_2} = \frac{R_2 + 1000 R_4}{1000 R_4 - R_1} \quad (4)$$

from (3) & (4)

$$\frac{R_2 + R_3 + R_4}{R_3 + R_4} = \frac{R_2 + 1000 R_4}{1000 R_4 - R_1}$$

↓

Put any value to get $V_0 = \frac{V_i}{1000}$



$$V_1 = i_1 R_1$$

$$V_3 = i_3 R_3$$

$$V_2 = i_2 R_2$$

$$V_4 = i_4 R_4$$

Applying KCL @ X

$$I = i_3 + i_4$$

KCL @ Y

$$i_2 = i_1 + i_3$$

KVL in loop 1

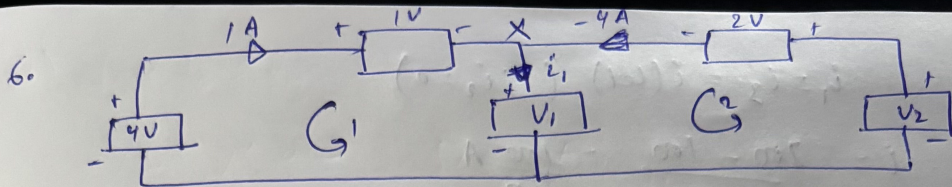
$$V - V_1 - V_2 = 0$$

KVL in loop 2

$$V_3 - V_4 + V_2 = 0$$

Solving all equations simultaneously, we get

$$V_4 = \frac{V R_2 R_4 + I R_1 R_2 R_4 + I R_1 R_3 R_4 + I R_2 R_3 R_4}{R_1 R_2 + R_1 R_3 + R_1 R_4 + R_2 R_3 + R_2 R_4}$$



KVL @ Loop G_1

$$V_1 + 1V + 4V = 0$$

$$\underline{V_1 = -5V}$$

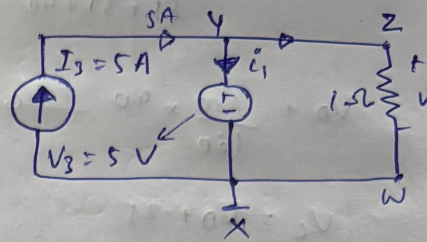
KVL @ Loop G_2

$$V_2 - 2 - 3 = 0$$

$$\underline{V_2 = 5V}$$

KCL @ X

$$i_1 = -4A + 1A = \underline{-3A}$$



Assume X to be grounded

$$V_X = 0V \quad V_Y = 5V$$

$$V_W = 0V \quad V_Z = 5V$$

$$V = V_{ZW} = 5V - 0V = 5V$$

$$\underline{V = 5V}$$

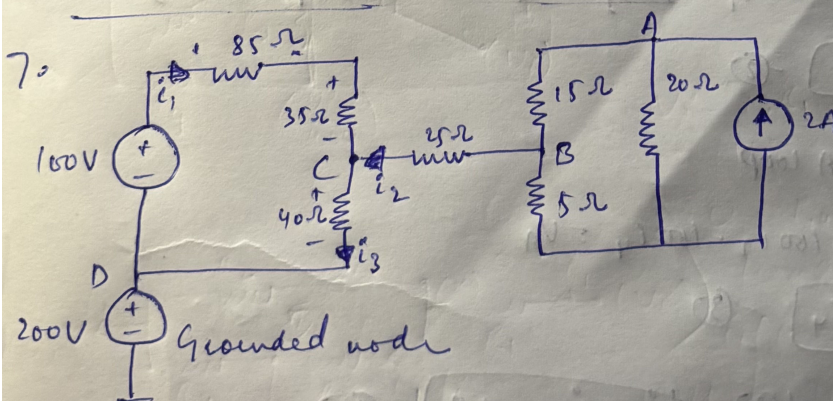
current through 1Ω resistor

$$\Rightarrow \frac{5}{1} = 5A$$

Applying KCL @ Y

$$i_1 + 5A = 5A$$

$$\boxed{i_1 = 0A}$$



Current will not flow through 25Ω resistor i.e. $i_2 = 0$
 because network of resistor on right of 25Ω is not grounded
 Hence no loop is formed with 25Ω resistor

Now $i_1 = i_3 = i$ (let) ($\because i_2 = 0$)

$$i = \frac{200 - 100}{85 + 35 + 40} = \frac{100}{160} \text{ A}$$

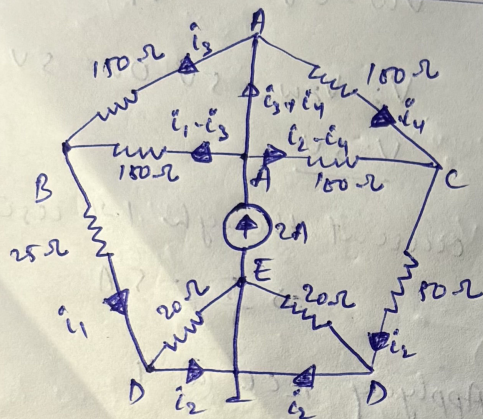
$$V_D + \frac{100 \times 40}{160} = V_C$$

$$V_C = V_D + 25$$

$$= 200 + 25$$

$$\boxed{V_C = 225 \text{ volt}}$$

8.



$$V_D = 0$$

$$\therefore V_E = 0 \Rightarrow \text{No current in } DE$$

Applying KCL @ grounded node

$$\boxed{i_1 + i_2 = 2 \text{ A}} \quad \text{--- (1)}$$

KVL in ABAA loop

$$V_A - 150 i_3 + 150 i_1 - 150 i_2 = V_A \Rightarrow 150 i_1 = 300 i_2$$

$$\boxed{i_1 = 2 i_2} \quad \text{--- (2)}$$

KVL in ACDA loop

$$V_A - 100 i_4 + 100 i_2 - 100 i_4 = V_A$$

$$\Rightarrow \boxed{i_2 = 2 i_4} \quad \text{--- (3)}$$

from (1) (2) and (3) $\boxed{i_2 + i_4 = 1 \text{ A}} \quad \text{--- (4)}$

KVL in ABACA loop

$$V_A - 150 i_3 + 150 i_1 - 150 i_3 + 100 i_2 + 100 i_4 + 100 i_4 = V_A$$

$$150 i_1 - 100 i_2 = 300 i_3 - 200 i_4$$

$$\boxed{3 i_1 - 2 i_2 = 6 i_3 - 4 i_4} \quad \text{--- (8)}$$

KVL in ABDDCA loop

$$V_A - 150 i_3 - 25 i_1 + 50 i_2 + 100 i_4 = V_A$$

$$\boxed{2 i_2 - i_1 = 6 i_3 - 4 i_4} \quad \text{--- (6)}$$

from (5) & (6)

$$4 i_1 = 4 i_2$$

$$\Rightarrow \boxed{i_1 = i_2} \quad \text{--- (7)}$$

$$\text{from (1) and (7)} \quad \boxed{i_1 = i_2 = 1 \text{ A}} \quad \text{--- (8)}$$

$$\text{(6) and (8)} \quad \boxed{6 i_3 - 4 i_4 = 1 \text{ A}} \quad \text{--- (9)}$$

$$\text{(4) and (9)} \quad \boxed{i_3 = i_4 = 0.5 \text{ A}} \quad \text{--- (10)}$$

Now $V_B = V_D + 25 i_1 = 25 \text{ V}$; $V_A = V_B + 150 i_3 = 100 \text{ V}$

$$V_C = 100 - 100 i_4 = 50 \text{ V};$$

$V_A = 100 \text{ V}$, $V_B = 25 \text{ V}$, $V_C = 50 \text{ V}$, $V_E = 0 \text{ volt}$ w.r.t $V_D = 0 \text{ (ground)}$

$$V_{AB} = 100 - 25 = 75 \text{ V}$$

$$V_{AC} = 100 - 50 = 50 \text{ V}$$

$$V_{BD} = 25 \text{ V}$$

$$V_{CD} = 80 \text{ V}$$

$$V_{DE} = 0 \text{ V}$$

$$V_{AE} = 100 \text{ V}$$